CSE 270 Sports Analytics

Basketball Individual performance

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NBA Individual Power Ratings

NBA Individual Power Ratings

2 main stages of the game.

- Attacking
- Defending

https://www.youtube.com/watch?v=wYjp2zoqQrs

NBA Individual Power Ratings

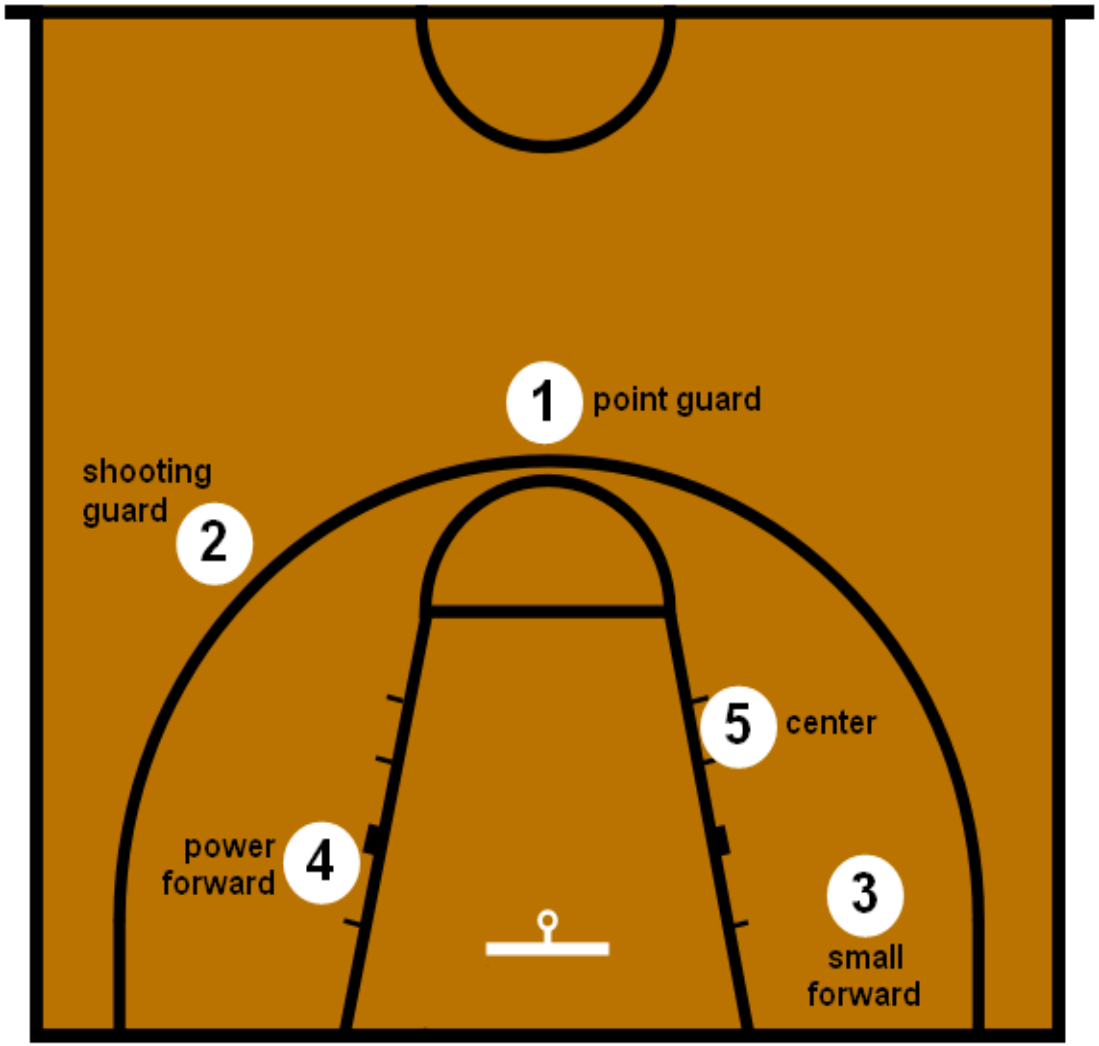
5 main positions on the field.

- Center - On offense, the center tries to score on close shots and rebound. But on defense, the center tries to block opponents' shots and rebound their misses.
- Power Forward - The power forward does many of the things a center does, playing near the basket while rebounding and defending taller players. But power forwards also take longer shots than centers.
- Small Forward - The small forward plays against small and large players. They roam all over on the court. Small forwards can score from long shots and close ones.
- Point Guard - The point guard runs the offense.The point guard defends the opponent's point guard and tries to steal the ball.
- Shooting Guard - The shooting guard can make shots from long distance and also is a good dribbler.

Let's watch a short video,

https://www.youtube.com/watch?v=4_4CymXARWQ

NBA Individual Power Ratings



NBA Individual Power Ratings

Defining attributes for each position.

- Center - The center is the tallest player on each team, playing near the basket.
- Power Forward - The power forward is very close to center player, but in this position they should also have good shooting attributes.
- Small Forward - The small forward should have good shooting efficiency in close and far distances.
- Point Guard - The point guard should have good defending characteristics.
- Shooting Guard - The shooting guard should have good dribbling attributes and shooting accuracy in long ranges.

What measures other than the height and shooting can be used to position the players on the field ?

NBA Individual Power Ratings

In game statistics

- STEALS
- TURNOVERS - Turnover occurs when a team loses possession of the ball to the opposing team before a player takes a shot at their team's basket.
  https://www.youtube.com/shorts/I4MDXOWrUmk
- REBOUNDS
  1. OFFENSIVE REBOUNDS - An offensive rebound occurs when offensive players regain possession of the ball after a missed shot.
  2. DEFENSIVE REBOUNDS - Defensive rebounds occur when defensive players gain possession of the ball after an offensive player misses the shot.
- BLOCKS - Block occurs when a defensive player legally deflects a field goal attempt from an offensive player to prevent a score.
- FOULS
  1. PERSONAL FOULS
  2. FLAGRANT FOULS
  3. TECHINCAL FOULS

NBA Individual Power Ratings

Dataset

```
str(nba_players)
```

```
## 'data.frame':    9708 obs. of  53 variables:
##  $ X     : int  14967 14968 14969 14970 14971 14972 14973 14974 14975 14976 ...
##  $ Year  : int  2001 2001 2001 2001 2001 2001 2001 2001 2001 2001 ...
##  $ Player: chr  "Mahmoud Abdul-Rauf" "Tariq Abdul-Wahad" "Shareef Abdur-Rahim" "Cory Alexander" ...
##  $ Pos   : chr  "PG" "SG" "SF" "PG" ...
##  $ Age   : int  31 26 24 27 23 23 23 25 24 30 ...
##  $ Tm    : chr  "VAN" "DEN" "VAN" "ORL" ...
##  $ G     : int  41 29 81 26 65 38 27 82 37 82 ...
##  $ GS    : int  0 12 81 0 24 6 18 82 2 36 ...
##  $ MP    : int  486 420 3241 227 1382 472 910 3129 288 1710 ...
##  $ PER   : num  16.7 5.8 19.1 6.3 11.6 6.8 14.2 22.9 12.1 8.7 ...
##  $ TS.   : num  0.514 0.438 0.549 0.407 0.484 0.404 0.519 0.61 0.429 0.455 ...
##  $ X3PAr : num  0.057 0.09 0.05 0.286 0.08 0.056 0.091 0.357 0.357 0.012 ...
##  $ FTr   : num  0.118 0.324 0.415 0.321 0.262 0.253 0.266 0.299 0.155 0.471 ...
##  $ ORB.  : num  1.2 3.6 6.2 0 3.5 4.9 2.8 3.7 1.6 4.9 ...
##  $ DRB.  : num  4.7 11.8 19.6 12.5 8.4 9.9 7.6 11.8 10.6 12.7 ...
##  $ TRB.  : num  2.9 7.6 13 6.1 5.9 7.5 5.1 7.8 6.1 8.7 ...
##  $ AST.  : num  32.7 8.1 14.3 23.5 8 6.9 8.6 20.4 34.6 7 ...
##  $ STL.  : num  1 1.7 1.5 3.6 1.7 1.7 1.7 2.1 2.4 0.8 ...
##  $ BLK.  : num  0.1 2.1 1.6 0 0.3 0.5 0.2 0.5 0 1.2 ...
##  $ TOV.  : num  9.1 21.1 13.2 28.1 10.5 9.6 10.9 12.1 18.2 14.8 ...
##  $ USG.  : num  26.8 17.1 24.6 17.3 23.2 21.1 24.3 24.4 17.2 21.7 ...
##  $ blanl : logi  NA NA NA NA NA NA ...
##  $ OWS   : num  0.7 -0.7 5.6 -0.5 0.3 -0.5 0.9 11.5 0.1 -1.3 ...
##  $ DWS   : num  0 0.3 2.9 0.3 0.5 0.4 0.1 2.2 0.2 1.4 ...
##  $ WS    : num  0.7 -0.4 8.5 -0.1 0.8 -0.1 0.9 13.7 0.3 0 ...
##  $ WS.48 : num  0.07 -0.049 0.126 -0.031 0.027 -0.013 0.048 0.211 0.055 0.001 ...
##  $ blank2: logi  NA NA NA NA NA NA ...
##  $ OBPM  : num  0.8 -5.9 1.3 -5.9 -1.8 -4.8 -0.3 6.5 -1.8 -5.1 ...
##  $ DBPM  : num  -5.3 0 0.9 -0.1 -2.5 -1.8 -2.9 -0.9 -2.7 -1.8 ...
##  $ BPM   : num  -4.5 -5.9 2.2 -6.1 -4.4 -6.6 -3.2 5.6 -4.5 -6.9 ...
##  $ VORP  : num  -0.3 -0.4 3.5 -0.2 -0.8 -0.5 -0.3 6 -0.2 -2.1 ...
##  $ FG    : int  120 43 604 18 239 62 177 628 30 237
```

NBA Individual Power Ratings

Let's select the performance metrics.

```
nba_players <- nba_players %>%
  select(Year, Player, Pos,
         G, MP, STL, TOV, ORB, DRB, PF, BLK, PTS, AST,
         FT, FTA, FG, FGA, X3P, X3PA, X2P, X2PA, ORB., DRB.,
         TRB., AST., STL., BLK., TOV., FG., X3P., FT.,) %>%
  arrange(Player, Year)
head(nba_players, 3)
```

```
##   Year      Player Pos  G   MP STL TOV ORB DRB  PF BLK PTS AST FT FTA  FG FGA
## 1 2001  A.C. Green  PF 82 1411  30  45 107 206 119   8 367  39 79 111 144 324
## 2 2001 A.J. Guyton  PG 33  630   9  24  10  26  35   5 198  64 15  18  78 192
## 3 2002 A.J. Guyton  PG 45  607  10  37  12  32  23   7 244  81 22  27  88 244
##   X3P X3PA X2P X2PA ORB. DRB. TRB. AST. STL. BLK. TOV.   FG.  X3P.   FT.
## 1   0    6 144  318  9.0 17.3 13.2  4.8  1.2  0.4 10.8 0.444 0.000 0.712
## 2  27   69  51  123  1.8  5.1  3.4 18.1  0.8  0.6 10.7 0.406 0.391 0.833
## 3  46  123  42  121  2.3  6.3  4.2 23.6  0.9  0.9 12.6 0.361 0.374 0.815
```

NBA Individual Power Ratings

Let's try to evaluate the player's performance based on the amount of time they spent on the field.
We will use the players' historical data to measure their performance for each Year, by taking 5 preceding Years only in order to avoid one Year stars. We will use success/failure approach by dividing the metrics of the players into successful and non-succesfull actions and then compare their success rate with their playing Minutes.

NBA Individual Power Ratings

Let's start with defending attributes.

Successful actions, when defending

- STEALS
- BLOCKS
- DEFENSIVE REBOUNDS

Non Successful actions, when defending

- PERSONAL FOULS
- TURNOVER

NBA Individual Power Ratings

Let's calculate the difference between the cumulative success actions and cumulative failure actions.

```
nba_players_defense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(STL_TOV = STL - TOV,
         STL_PF = STL - PF,
         DRB_TOV = DRB - TOV,
         BLK_TOV = BLK - TOV,
         BLK_PF = BLK - PF) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_defense$MP))
}

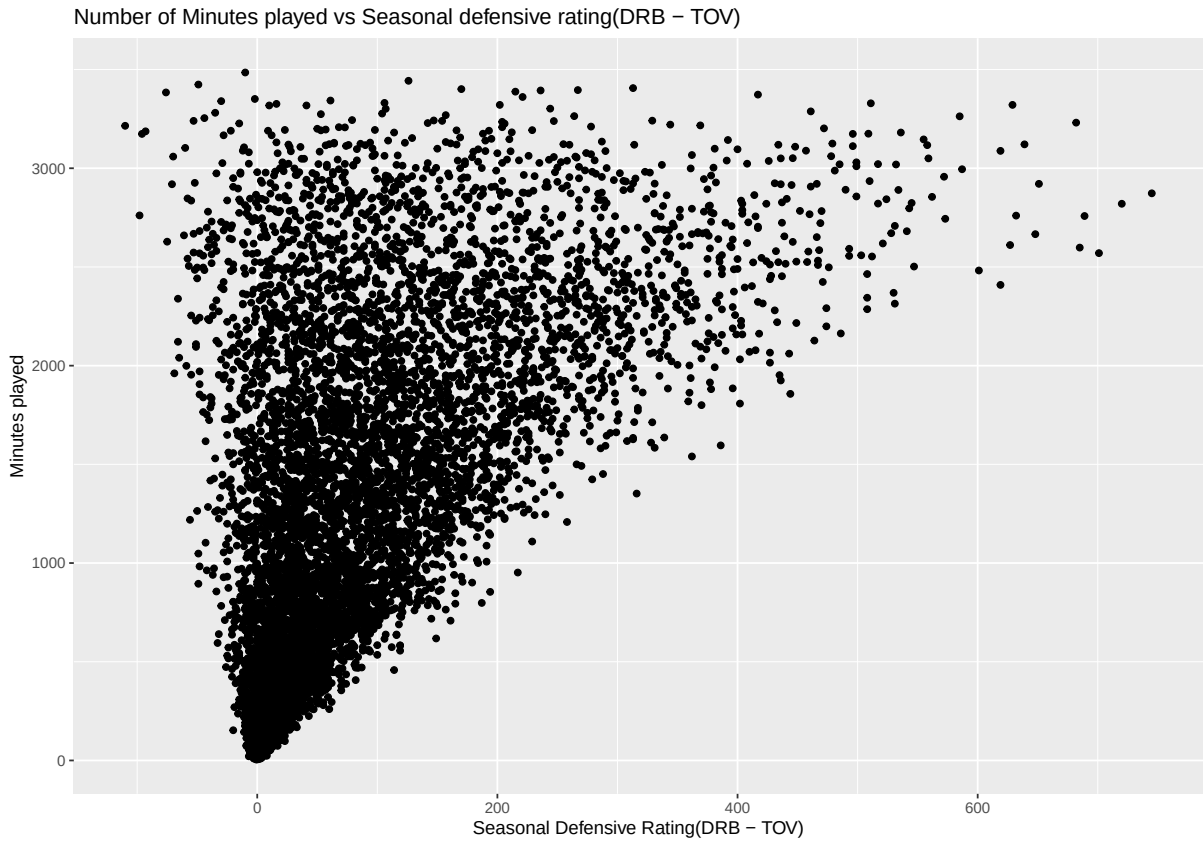
nba_players_defense %>%
  select(STL_TOV, STL_PF, DRB_TOV, BLK_TOV, BLK_PF) %>%
  apply(2,cor_with_MP)
```

```
##    STL_TOV     STL_PF    DRB_TOV    BLK_TOV     BLK_PF
## -0.6511430 -0.6257446  0.5812748 -0.6796643 -0.7917445
```

NBA Individual Power Ratings

The metric with the highest positive correlation rate is the Seasonal difference between the seasonal number of defensive rebounds made and number of turnovers made.

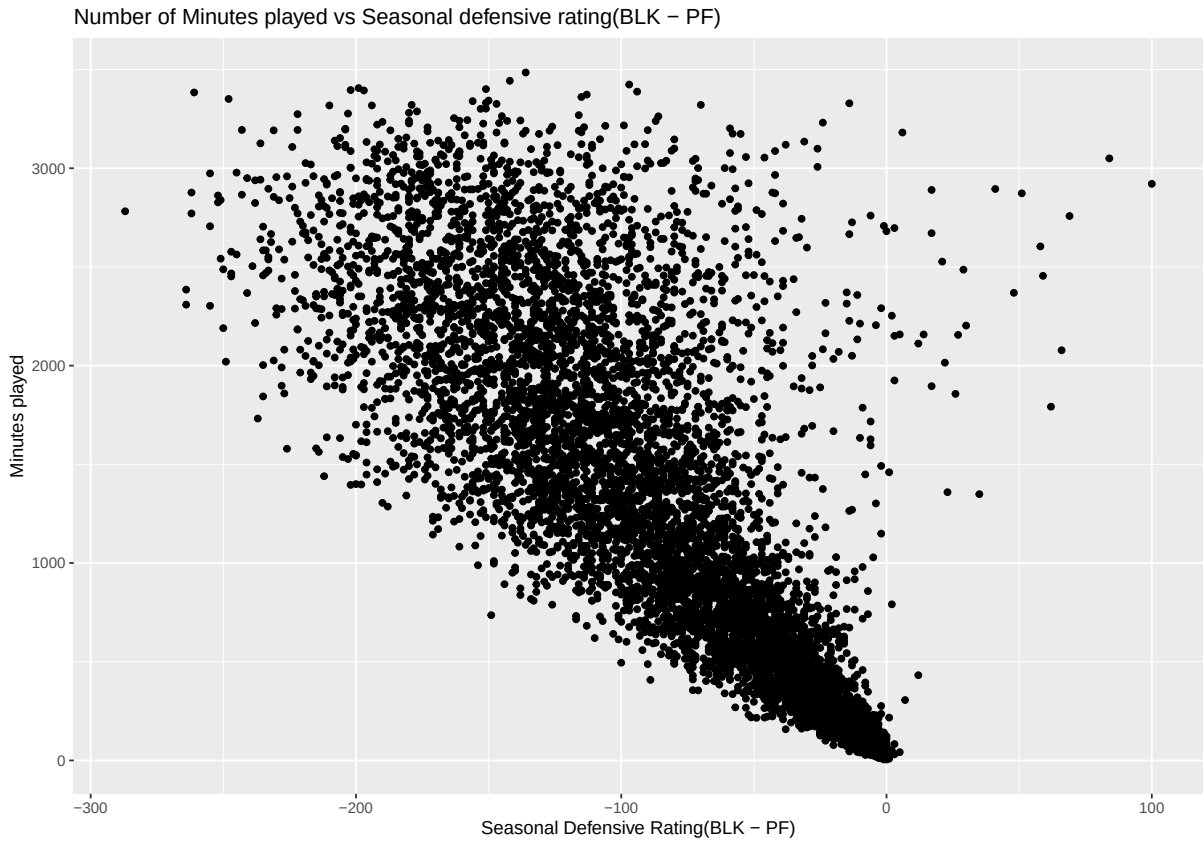
```
ggplot(nba_players_defense, aes(x = DRB_TOV, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(DRB - TOV)") +
  xlab("Seasonal Defensive Rating(DRB - TOV)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

The metric with the highest negative correlation rate is the difference between the cumulative number of blocks made and cumulative number of turnovers made.

```
ggplot(nba_players_defense, aes(x = BLK_PF, y = (MP))) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal defensive rating(BLK - PF)") +
  xlab("Seasonal Defensive Rating(BLK - PF)") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

Let's fit a model with the defensive power ratings.

```
summary(lm(MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF ), data = nba_players_defense))
```

```
##
## Call:
## lm(formula = MP ~ (STL_TOV + STL_PF + DRB_TOV + BLK_PF), data = nba_players_defense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2468.96  -176.08   -78.07   161.59  2172.64
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 160.05456    7.17817   22.30   <2e-16 ***
## STL_TOV      -8.15592    0.12100  -67.41   <2e-16 ***
## STL_PF        7.29159    0.14525   50.20   <2e-16 ***
## DRB_TOV       4.01781    0.05039   79.73   <2e-16 ***
## BLK_PF      -11.81019    0.11231 -105.16   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 352.7 on 7852 degrees of freedom
## Multiple R-squared:  0.8413, Adjusted R-squared:  0.8412
## F-statistic: 1.04e+04 on 4 and 7852 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

- Majority of the players make many fouls, but still play relatively often if they make more steals.
- Having more turnovers mostly causes less playing practice.
- Having more defensive rebounds and steals can improve player's playing practice.

NBA Individual Power Ratings

For the attacking attributes we will calculate the difference between the attempts made for successful actions and actual successful actions.

Successful actions, when attacking

- FIELD GOALS
- 3 POINT GOALS
- FREE THROWS
- OFFENSIVE REBOUNDS
- POINTS SCORED
- ASSISTS MADE

Non Successful actions, when attacking

- FIELD GOALS ATTEMPTS
- 3 POINT ATTEMPTS
- FREE THROW ATTEMPTS

NBA Individual Power Ratings

```
nba_players_offense <- nba_players %>%
  arrange(Player, Year)  %>%
  group_by(Player) %>%
  mutate(FGA_FG = FGA - FG,
         X3PA_X3P = X3PA - X3P,
         FTA_FT = FTA - FT) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual Power Ratings

Let's check if the Seasonal metrics are correlated to the amount of time the players spent on the field.

```
cor_with_MP <- function(x){
  return(cor(x, nba_players_offense$MP))
}

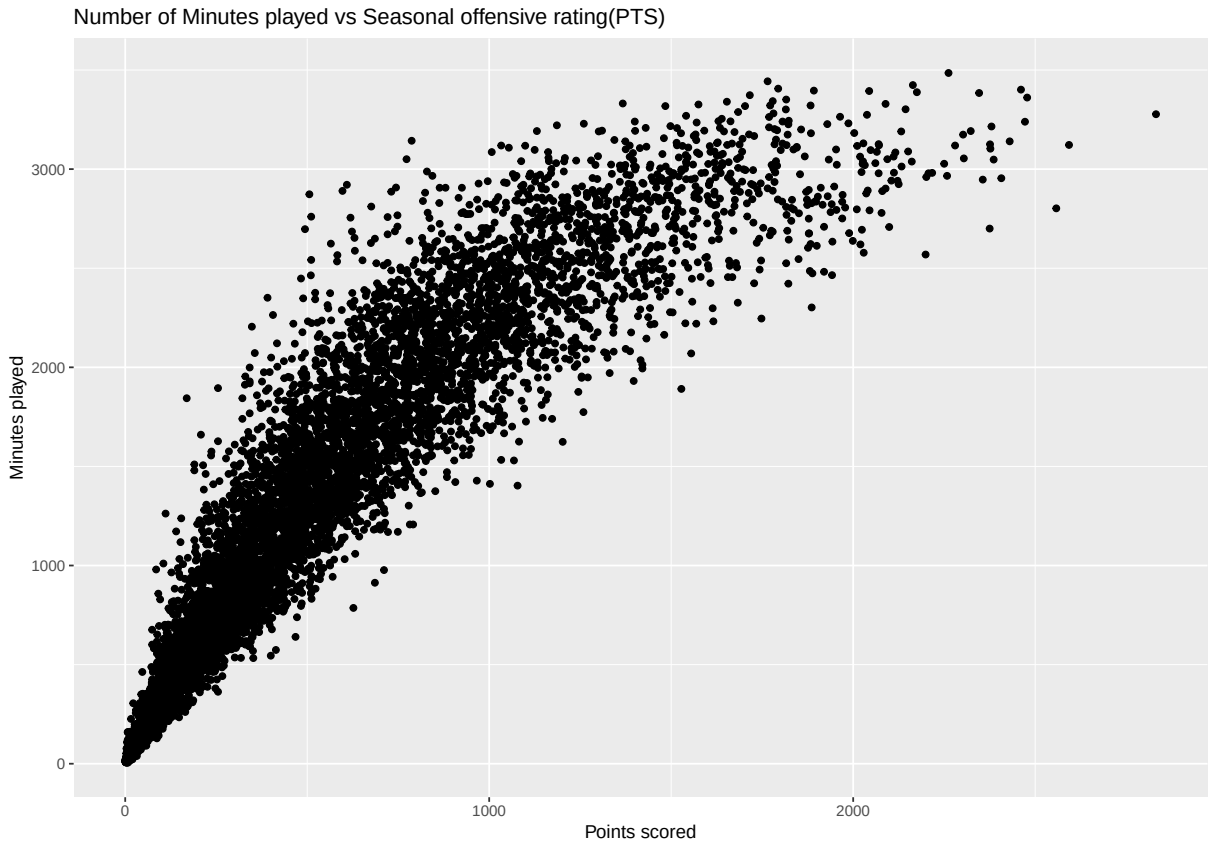
nba_players_offense %>%
  select(FGA_FG, X3PA_X3P, FTA_FT, AST, PTS) %>%
  apply(2,cor_with_MP)
```

```
##    FGA_FG  X3PA_X3P    FTA_FT       AST       PTS
## 0.9198331 0.6245125 0.6945966 0.7237956 0.9217370
```

NBA Individual Power Ratings

The Seasonal metric with the highest correlation is the difference between the number of points scored by the player.

```
ggplot(nba_players_offense,aes(x = PTS, y = MP)) +
  geom_point() +
  ggtitle("Number of Minutes played vs Seasonal offensive rating(PTS)") +
  xlab("Points scored") +
  ylab("Minutes played")
```



NBA Individual Power Ratings

```
summary(lm(MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MP ~ (PTS + AST + FGA_FG + X3PA_X3P + FTA_FT), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2343.50  -213.33   -51.15   180.09  1428.56
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 284.94753    5.78883  49.224  < 2e-16 ***
## PTS           0.61928    0.03855  16.065  < 2e-16 ***
## AST           0.77363    0.03999  19.345  < 2e-16 ***
## FGA_FG        1.91642    0.08863  21.622  < 2e-16 ***
## X3PA_X3P      0.23385    0.07714   3.031  0.00244 **
## FTA_FT        3.04911    0.18524  16.460  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 318.2 on 7851 degrees of freedom
## Multiple R-squared:  0.8708, Adjusted R-squared:  0.8707
## F-statistic: 1.058e+04 on 5 and 7851 DF,  p-value: < 2.2e-16
```

NBA Individual Power Ratings

- Scoring more points/assisting = playing more
- Attempting more to score a field goal or a field throw.